TOXICITY LLM CLASSIFICATION

LLM-Based Toxicity Prediction for Drug Discovery with SMILES and ChemBERTa

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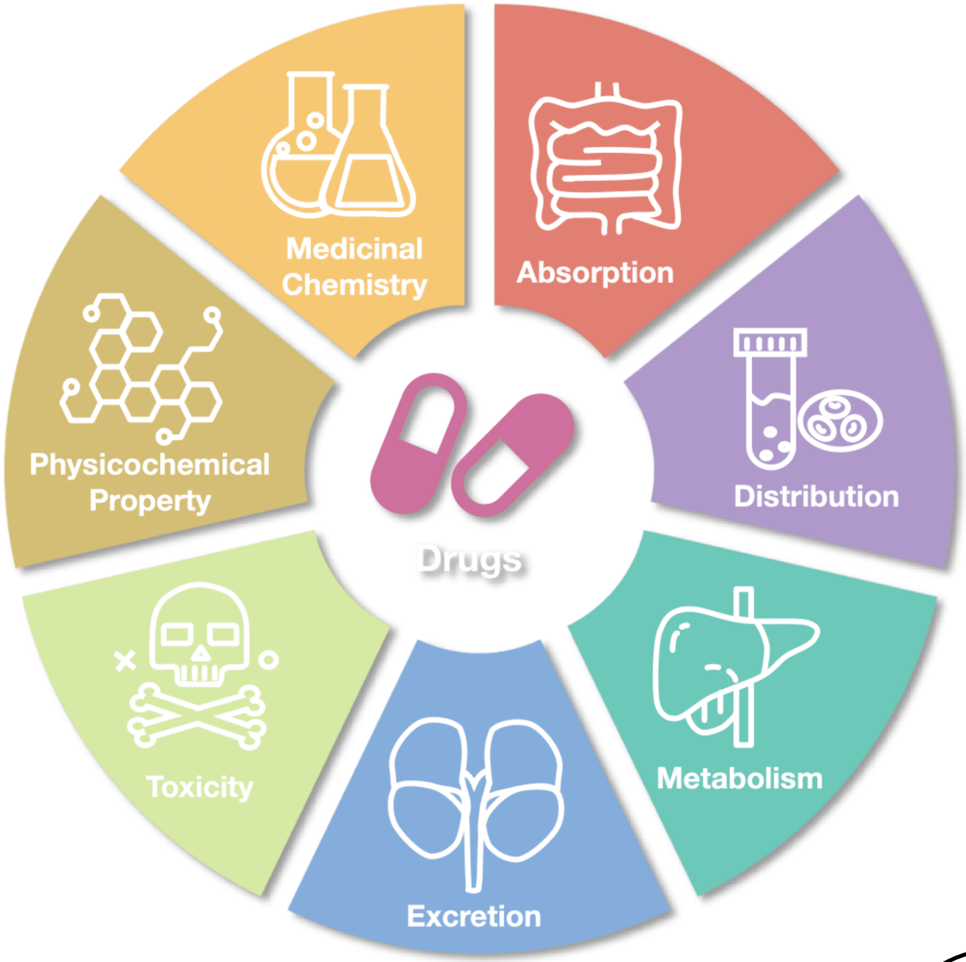
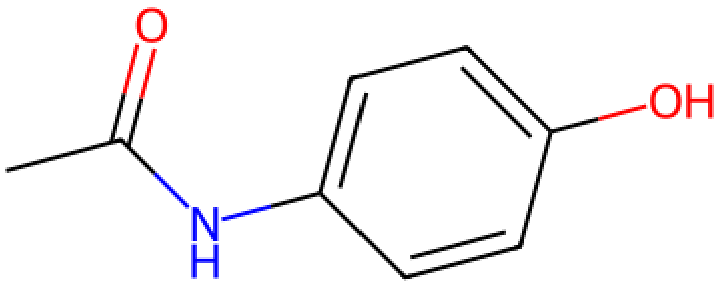
Drug Discovery Pipeline: ADMET

- In Silico Toxicity Prediction
- In Vitro Toxicity Assays
- In Vivo Toxicity Studies
- Safety Pharmacology
- Toxicokinetics
- Risk Assessment & Optimization



Small molecule Toxicity classification

- Small molecule: organic compounds with <900 daltons
- Evolved over last two decades
- Key: powerful computational resources, good quality data
- Commonly used representations
  - SMILES embeddings
  - Morgan Fingerprint

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'CC(=O)NC1=CC=C(C=C1)O'
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Clintox dataset: FDA Approval result and toxicity

Dataset description: Clintox

- Total 1480 datapoints
- Drugs that have succeeded (approved by FDA) /failed clinical trials for toxicity reasons
- Strong relationship between two labels; FDA_APPROVED & CT_TOX

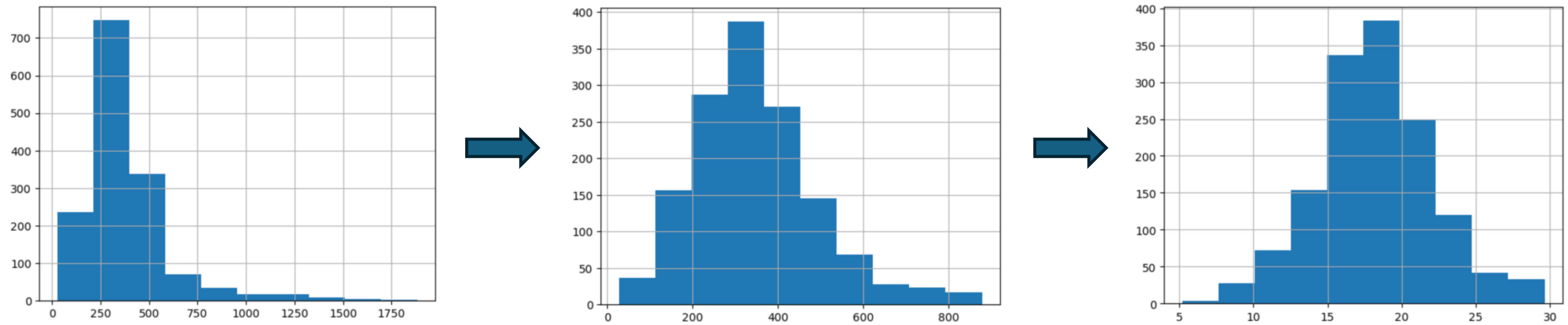
	smiles	FDA_APPROVED	CT_TOX
0	*C(=O)[C@H](CCCCNC(=O)OCCOC)NC(=O)OCCOC	1	0
1	[C@@H]1([C@@H]([C@@H]([C@H]([C@@H]([C@@H]1Cl)C...	1	0
2	[C@H]([C@@H]([C@@H](C(=O)[O-])O)O)([C@H](C(=O)...	1	0
3	[H]/[NH+]=C(/C1=CC(=O)/C(=C\C=c2ccc(=C([NH3+])...	1	0
4	[H]/[NH+]=C(\N)/c1ccc(cc1)OCCCCCOc2ccc(cc2)/C(...	1	0
5	[N+](=O)([O-])[O-]	1	0

Previous research: Mol-BERT

Model	Mol-BERT	ChemBERTa
Input	SMILES	
Pretrained Data	4M molecules	77M molecules
Tokenizer	Atom-level or substructure-level	Byte-level (BPE)
Base Model Engine	BERT	RoBERTa
Characteristic	Chemistry knowledge embedded	Large-scale training, more general
Main Purpose	Emphasize chemical meaning	Leverage LLM style training

Data Preprocessing

- Removal of NA containing datapoints (5 datapoints lost)
- Small molecules (<900 daltons) only (60 datapoints lost)
- Regularization of the molecular weight
  - Common method: log/sqrt/min-max scaling
  - In this dataset: log & sqrt



Data Preprocessing

- Removal of invalid SMILES
- Removal of salt, hydrogen acceptor from SMILES
- Canonicalization of SMILES
- Tokenized feature extraction from SMILES with ChemBERTa

	smiles
1	C(=O)[C@H](CCCCNC(=O)OCCOC)NC(=O)OCCOC
2	Cl[C@H]1[C@H](Cl)[C@@H](Cl)[C@@H](Cl)[C@H](C...
3	O=C([O-])[C@@H](O)[C@@H](O)[C@H](O)[C@@H](...
4	[H]/[NH+]=C(\N)C1=CC(=O)/C(=C\C=c2ccc(=C(N)[...
5	[H]/[NH+]=C(\N)c1ccc(OCCCCCOc2ccc(/C(N)=[NH+...

ChemBERTa →

tokenized_features
[-1.20778829e-01 -5.82895458e-01 -6.78249002e-...
[-2.02718028e-03 -3.41230929e-01 -8.83470029e-...
[-3.94689590e-01 -4.38813806e-01 -2.99492121e-...
[0.8228974 0.18262821 0.23519734 -0.0645721...
[0.40231556 -0.1371812 0.20072028 0.1538573...

Model baseline: Random Forest

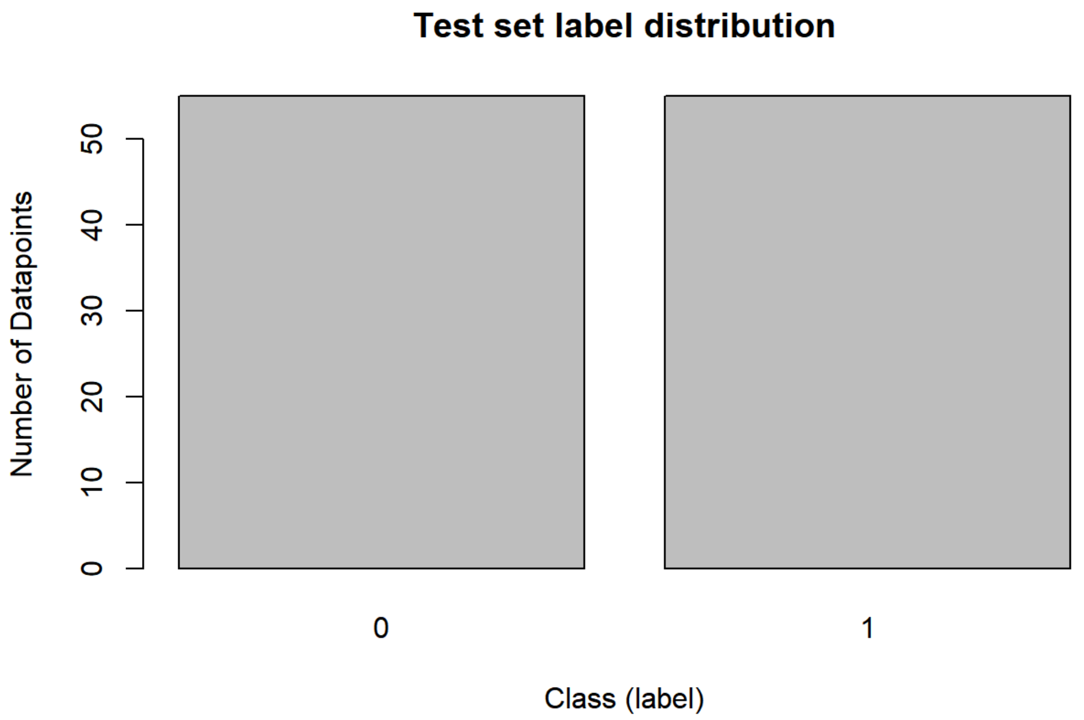
- Training dataset input features (dimension)
  - Fingerprint (1, 1024)
  - Tokenized features (1, 384)
  - Descriptors as Float64

fingerprint	tokenized_features
[0 1 0 ... 0 0 0]	[-1.20778829e-01 -5.82895458e-01 -6.78249002e-...
[0 0 0 ... 0 0 0]	[-2.02718028e-03 -3.41230929e-01 -8.83470029e-...
[0 1 0 ... 0 0 0]	[-3.94689590e-01 -4.38813806e-01 -2.99492121e-...
[0 0 0 ... 0 0 0]	[0.8228974 0.18262821 0.23519734 -0.0645721...
[0 0 0 ... 0 0 0]	[0.40231556 -0.1371812 0.20072028 0.1538573...

molecular_weight	molecular_logp
18.285759	0.46940
17.053797	3.64440
14.426434	-6.07020
16.803184	-3.15860
18.505215	-0.75650

Test Dataset Balancing

- Class unbalance causes difficulty on evaluating the model
- The test set label class should be balanced



Clintox classification result

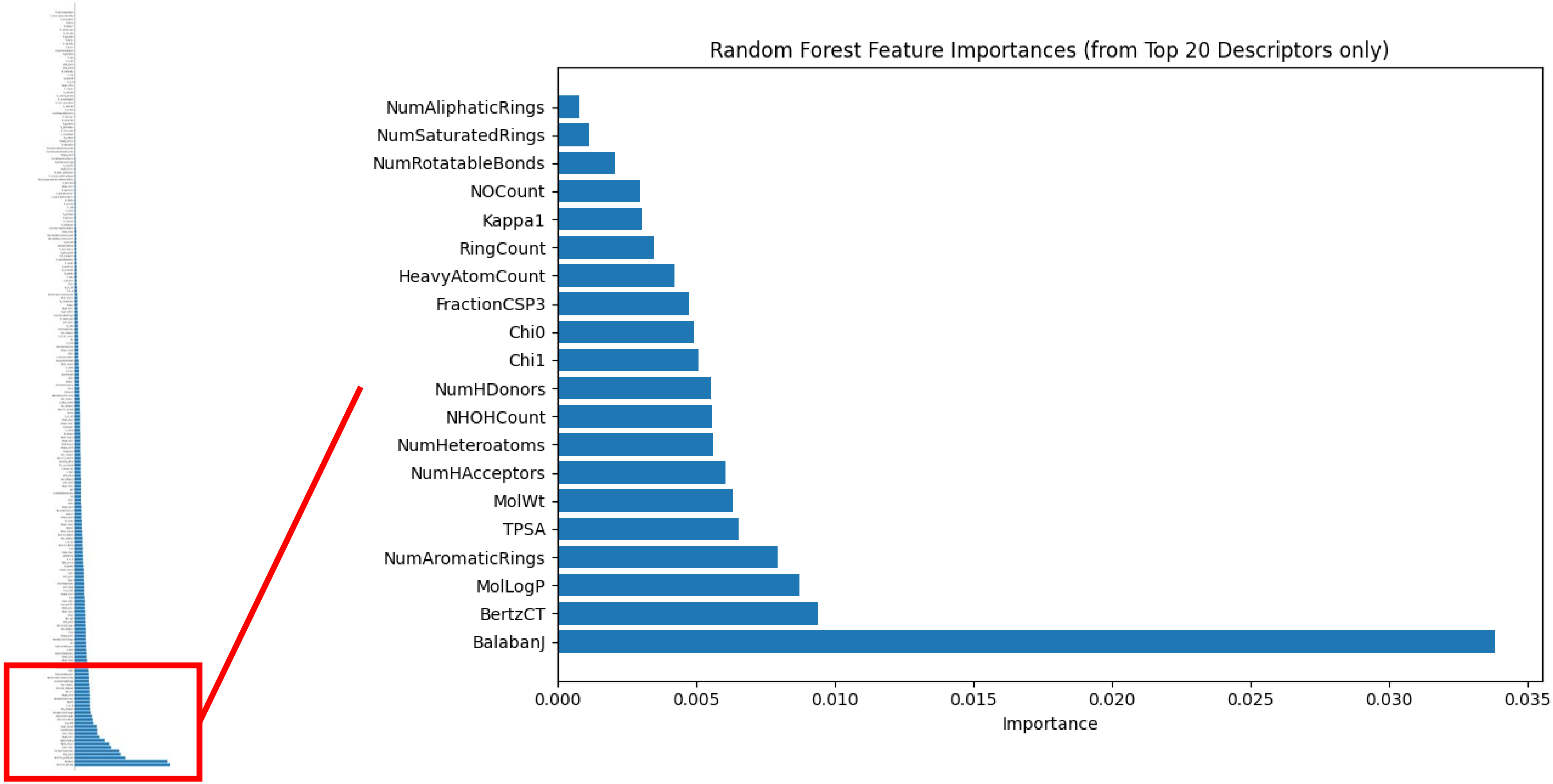
Experiment condition		Accuracy	Precision	Recall	F1-score	ROC-AUC
MolBERT		-	-	-	-	0.9230
Pure RandomForest	Random train-test div	0.9144	-	0	-	0
	Class balanced test div	0.5	-	0	-	0
Weighed RandomForest	Class weight 1:5	0.9020	0.4000	0.4166	0.4081	-
	Class weight 1:10	0.8885	0.3513	0.5909	0.4406	-
Downsampled RandomForest	Random test, without molinfo	0.2703	0.0867	**0.7895**	0.1565	0.5056
	Balanced test, without molinfo	0.5963	0.5766	0.7248	0.6411	0.5963
	Balanced test, with molinfo	0.7661	0.7685	0.7615	0.7650	**0.7661**
Upsampled RandomForest	Class balanced test div	0.5	-	0	-	0

Performance with molecular features

Comparative evaluation of Random Forest model performance to determine the most influential individual features and the most effective combinations of molecular descriptors

Model	Accuracy	Precision	Recall	F1_Score
Molecular Weights + logP	0.7661	0.7636	0.7706	0.7671
Tokenized features	0.5138	0.5133	0.5321	0.5226
Tokenized features + Molecular Weights + logP	0.5826	0.5542	0.8440	0.6641
Tokenized features + 20 descriptors	0.9286	0.8643	0.9300	0.8959
Tokenized features + all RDKIT descriptors	0.9328	0.8749	**0.9344**	0.9037

Rdkit descriptor importance as feature

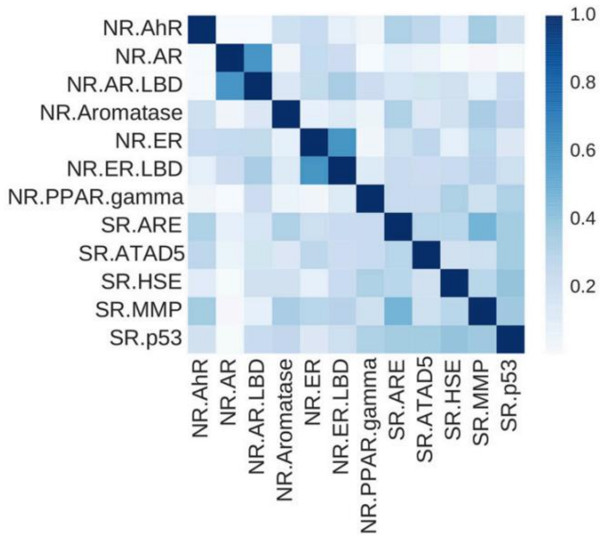


TOX21 dataset: Multi-toxicology assays for new toxicity assess method search

Dataset description: Tox21

Feature Name	Description
smiles	SMILES string representing the chemical compound structure.
NR-AR	Activation of the **androgen receptor (AR)** — associated with reproductive toxicity and hormone disruption.
NR-AR-LBD	Binding to the **ligand-binding domain** of the AR — measures physical interaction with AR.
NR-AhR	Activation of the **aryl hydrocarbon receptor (AhR)** — linked to xenobiotic metabolism and immune modulation.
NR-Aromatase	Inhibition of **aromatase enzyme** — affects estrogen synthesis; relevant in reproductive toxicity.
NR-ER	Activation of the **estrogen receptor (ER)** — involved in hormone regulation and endocrine disruption.
NR-ER-LBD	Binding to the **ligand-binding domain of ER** — direct physical binding assay.
NR-PPAR-gamma	Activation of the **peroxisome proliferator-activated receptor gamma** — involved in lipid metabolism and adipogenesis.
SR-ARE	Activation of the **antioxidant response element (ARE)** pathway — indicates oxidative stress potential.
SR-ATAD5	Activation of **ATAD5**-linked pathway — associated with DNA replication stress and repair.
SR-HSE	Activation of **heat shock element (HSE)** pathway — cellular response to protein damage and heat shock.
SR-MMP	Modulation of **mitochondrial membrane potential (MMP)** — an early indicator of apoptosis or mitochondrial toxicity.
SR-p53	Activation of **p53 tumor suppressor protein** — signals genotoxic stress or DNA damage.

- **Total 1907 datapoints**
- **Chemical compounds evaluated across 12 different toxicological assays to develop new methods for toxicity testing**

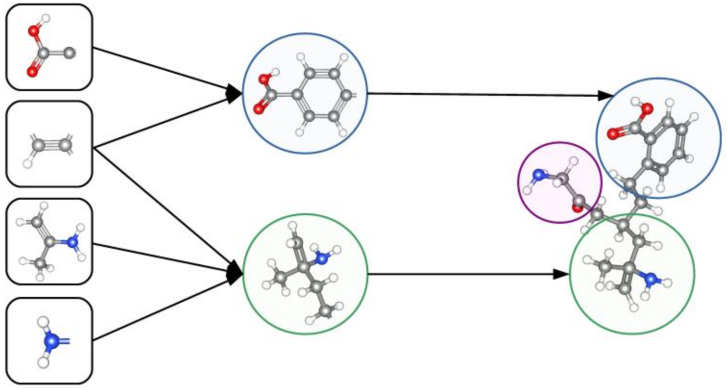


Previous research: DeepTox project

- Dataset: Tox21
  - 801 "dense features" that represent chemical descriptors, such as molecular weight, solubility or surface area
  - 272,776 "sparse features" that represent chemical substructures
- Best performing method in Tox21 Data Challenge
- Dense features + sparse features -> DNN -> random forest

Sparse Feature
[0, 1, 0, 0, 0, 1, 0, ...]

Dense Feature
[0.45, 0.82, 0.01]



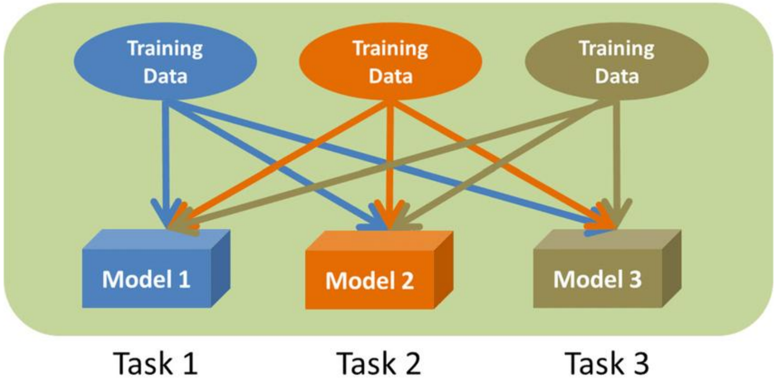
MTR vs MLM

ex) ChemBERTa-10M-MTR: ChemBERTa pretrained as multi-task regression with 10M molecules

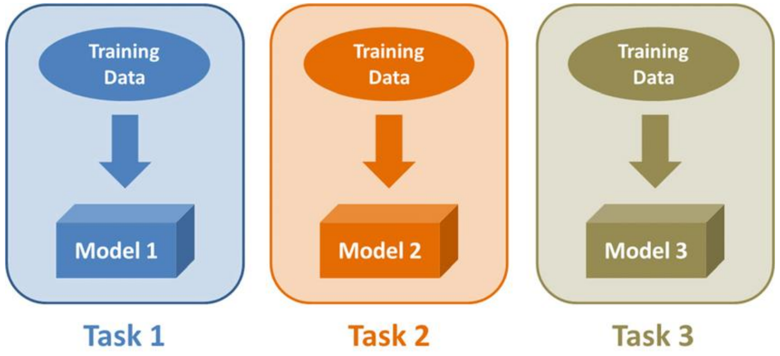
Feature	MTR (Multi-Task Regression)	MLM (Masked Language Modeling)
Goal	Predict multiple molecular properties simultaneously	Predict masked tokens in SMILES strings
Type	Supervised, regression-based	Self-supervised, language-based
Training signal	Use labeled molecular data with known properties	Use large, unlabeled SMILES datasets
Domain awareness	Highly domain-specific, focused on chemical properties	Less domain-specific, focused on SMILES syntax
Complexity	Require curated multi-property datasets	Only requires raw SMILES strings
Data efficiency	More data-efficient for downstream tasks	Needs more data to generate well

MTL vs STL

- **Multi-task learning (MTL)**
  - All toxicity targets are predicted simultaneously by a single model
  - The model learns shared representations across tasks
  - Captures the relationships between different toxicity endpoints



- **Single-task learning (STL)**
  - Each toxicity task (target) is treated independently
  - A separate model is trained for each task
  - The model performance is evaluated separately for each task



Tox21 multitask classification result

Task	BM (MTL)	Multi-task Classification			Single-task Classification		
	DeepTox	10M-MTR	77M-MTR	77M-MLM	10M-MTR	77M-MTR	77M-MLM
NR-AR	0.7445	0.7722	0.7536	0.7384	0.7752	0.7830	0.6814
NR-AR-LBD	0.7018	0.7726	0.7673	0.8024	0.7509	0.8181	0.7739
NR-AhR	0.9029	0.8009	0.7951	0.7599	0.7759	0.7728	0.6506
NR-Aromatase	0.7964	0.6739	0.6765	0.6626	0.6831	0.6609	0.5914
NR-ER	0.7735	0.6910	0.6789	0.6709	0.6686	0.6753	0.6073
NR-ER-LBD	0.7545	0.7217	0.7140	0.6776	0.7231	0.6733	0.6446
NR-PPAR-gamma	0.7618	0.6478	0.7142	0.7265	0.7176	0.6654	0.4962
SR-ARE	0.7771	0.6716	0.6491	0.6039	0.6737	0.6509	0.6099
SR-ATAD5	0.8152	0.6882	0.6979	0.6671	0.6523	0.6951	0.6250
SR-HSE	0.7814	0.7529	0.7293	0.7009	0.6711	0.6844	0.6180
SR-MMP	0.9246	0.8127	0.8008	0.7794	0.7402	0.7291	0.6675
SR-p53	0.8029	0.7575	0.7414	0.6897	0.7653	0.7580	0.6661

Discussion and Future Study Plans

Discussion: Clintox

- Downsampling helps because Clintox data often overfits the model
- Morgan Fingerprint has less impact to the ML model compared to tokenized SMILES
- Adding molecular information features can significantly improve model performance

Discussion: Tox21

- Multi-task learning (MTL) outperformed single-task learning (STL) across all metrics
- MTL helped mitigate class imbalance problem by sharing useful information across tasks
- MTR gives better performance than MLM in STL
- Large pretraining dataset size does not assure better performance
- Tokenized SMILES (ChemBERTa embeddings) provided rich, contextual representations of molecules

Discussion

- Balanced dataset is critical in drug discovery field
- Pretrained, Finetuned BERT (ChemBERTa) is useful for SMILES classification task in case of clintox dataset but less useful in tox21 dataset (worse performance relative to the benchmark; deeptox)
- SMILES as natural language can be reflected into ML (Machine Learning) by using LLM (Large Language Model) tokenizer; but this does not assure dramatic increase of the model performance
- ML with LLM can be used as a finalize approver in drug discovery pipeline (Clintox), or be used as front defender for toxic materials in drug discovery (Tox21)

Future study

- Undersampling vs Oversampling vs Data augmentation
- Best dataset combination for toxicity prediction (Clintox/Tox21/dense/sparse …)
- LLM with GNN: graph-based toxicity prediction (classification) using tokenized SMILES
- Transfer Learning with Domain-Specific Models
- Multi-modal Toxicity Prediction Models
- Explainable Toxicity Prediction Models

References

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THANK YOU

Any Questions?

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